

Curriculum Vitae

Kishal Tandel

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Education

- August, 2023 – present **BS-MS in Physical Sciences**, Indian Institute of Science Education and Research, Kolkata, India
CGPA : 8.75/10 (after 3rd Semester)
Relevant Courses : Classical Mechanics, Quantum Mechanics, Electromagnetism, Modern Physics, Analysis
- 2023 **ISC**, METAS M.C.D. School of S.D.A. , Surat, Gujarat, India
Class 12 - 93.25%, Indian School Certificate (ISC)
- 2021 **ICSE**, METAS M.C.D. School of S.D.A. , Surat, Gujarat, India
Class 10 - 95.40%, Indian Certificate of Secondary Examination (ICSE)

Research Interests

Quantum optics, Quantum gravity

Papers

- April 14, 2025 **One-Dimensional Quantum Tunneling**
Solved one-dimensional time independent Schrödinger equation for all energy regimes, derived the transmission and reflection coefficients and proved their continuity explicitly at $E = V_0$, plotted the coefficients using Python and compared the results with classical mechanics qualitatively.

Projects

- April, 2025 **Quantum Snake**
Developed a Python-based Quantum Snake game inspired by my work on one-dimensional tunneling, featuring a classic snake, constant potential barriers and a quantum food particle, whose motion is restricted in one dimension at a time.

Skills

- Programming Python, C, C++, Java, $\text{\LaTeX}$, Typst, HTML, CSS, JavaScript
- Language English, Hindi, Gujarati